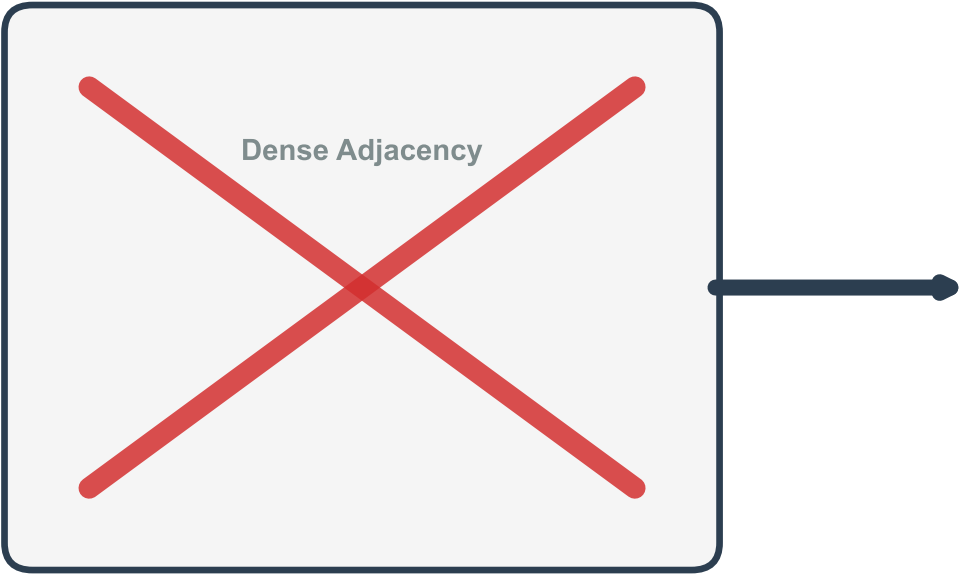
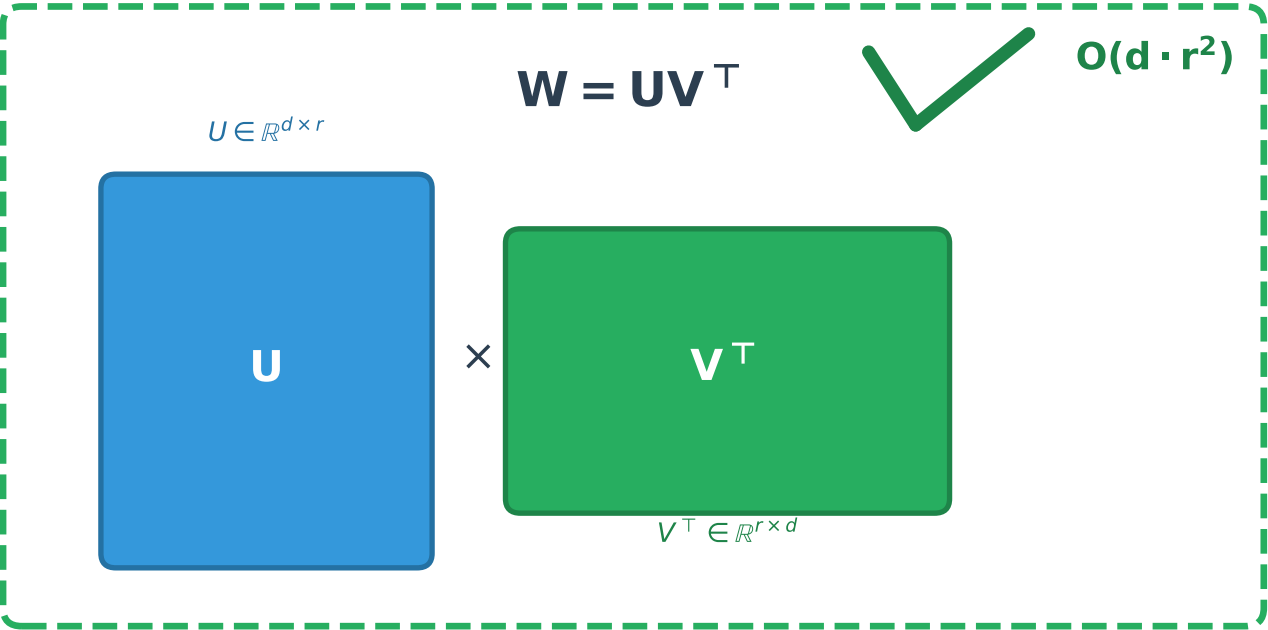


Low-Rank Factorization Breaks the Cubic Barrier in Causal Discovery

$O(d^3)$ Matrix Exponential



d^2 parameters



$2dr$ parameters ($r \ll d$)

Three Key Components



Adaptive Rank Selection

(Auto r via SVD knee)



Multi-Scale Decomposition

(Coarse + mid + fine)



Uncertainty Quantification

(Bootstrap ensembles)